

G3 User Stories

AMS Intake Platform - Algorithmic Intelligence Engine

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Group 3 — Algorithmic Intelligence Engine
<https://github.com/hodncgi/AMSSchatBot>

1. Introduction

This document lists end-user stories for the Group 3 AMS intake platform. The onboarding chatbot is available on the Onboarding page only. It answers questions using the G3 backend: POST /chat, POST /change-risk, POST /nbq/next, and GET /health. The deliverable is the standalone AMSChatBot project with no external site merge.

2. User Personas

- **New employee:** Uses the Onboarding page chatbot to learn CGI processes and tools.
- **Intake consultant:** Needs NBQ and change risk capabilities; asks the chatbot to explain scoring and prioritization.
- **AMS manager:** Reviews demos; needs a clear view of assistant status and answer mode without technical detail.
- **Knowledge manager:** Maintains onboarding documents in the relational database for chatbot accuracy.
- **Group 2 developer:** Will integrate REST APIs and database-backed knowledge into the wider AMS portal.
- **Project referent:** Validates scope and risk governance boundaries.

3. Product Scope

- Chatbot location: Onboarding tab only (not Dashboard, Documentation, or Manager views).
- Delivery: core POC Weeks 1–8; Weeks 9–11 for feedback, G2 integration, and polish.
- Week 6 milestone: receive Group 2 API routes (contact pending as of Week 2).
- Chatbot intelligence: connected to G3 FastAPI routes for chat, change risk, NBQ, and health.
- Knowledge (POC): curated files (knowledge_base.py, catalog.json); US-15 and US-16 target Group 1 relational database in v2.
- No external site merge: standalone React and FastAPI project (AMSChatBot repository).
- Change risk: decision-support score; AMS leads retain final sign-off.
- AI: local Mistral via LM Studio; RAG from onboarding knowledge (file-based today, database-backed planned).
- Trust and transparency: sources visible (US-13), uncertainty stated clearly (US-14).

4. User Stories

Format: As a [persona], I want [goal], so that [benefit]. Acceptance criteria define when the story is done.

US-01 - New employee

As a new CGI employee on the onboarding page, I want to ask questions in natural language, so that I understand processes and tools without searching multiple documents.

Priority: Must

Acceptance criteria:

- Chatbot is available on the Onboarding page only.
- I can type a question and receive an answer in English.
- The answer cites CGI documentation sources when available.

US-02 - New employee

As a new employee, I want suggested starter questions, so that I know what the assistant can help with.

Priority: Should

Acceptance criteria:

- At least four suggested questions are shown in the UI.
- Clicking a suggestion sends it to the chatbot.

US-03 - Intake consultant

As an intake consultant, I want the chatbot to explain change risk scoring, so that I understand how criticality and test days affect the score.

Priority: Must

Acceptance criteria:

- Chatbot can explain the change risk formula (0-100, drivers).
- Chatbot uses backend knowledge of POST /change-risk.
- User can ask "What is the risk for high criticality with 5 test days?" and get a coherent answer.

US-04 - Intake consultant

As an intake consultant, I want the chatbot to explain Next-Best Question logic, so that I know why a question is prioritized for a client sector.

Priority: Must

Acceptance criteria:

- Chatbot can describe NBQ weighting (e.g. Healthcare + security).
- Chatbot references POST /nbq/next behaviour.
- Explanation includes why a question matters for a sector.

US-05 - Intake consultant

As an intake consultant, I want to get the next best question for a client profile via the platform, so that I focus on the most relevant intake topic.

Priority: Must

Acceptance criteria:

- POST /nbq/next returns id, text, weight, and why.
- Healthcare + security scenario returns the security question with boosted weight.

US-06 - Intake consultant

As an intake consultant, I want to compute a change risk score from criticality and test days, so that I document risk in the intake dossier.

Priority: Must

Acceptance criteria:

- POST /change-risk returns score and drivers[].
- Score is between 0 and 100.
- Drivers explain how the score was calculated.

US-07 - AMS manager

As an AMS manager, I want to see the AI pipeline status and answer mode, so that I can understand whether the assistant is using the full AI service or a fallback mode.

Priority: Should

Acceptance criteria:

- The onboarding page shows a clear status for the assistant (full AI service vs fallback).
- Each answer indicates which mode was used, without requiring technical knowledge.
- When fallback is active, it is obvious that answers may rely on stored documentation only.
- A manager can assess assistant reliability during a demo without opening developer tools.

US-08 - Developer / Group 2

As a Group 2 developer, I want documented REST APIs for NBQ, change risk, and chat, so that I can integrate the G3 engine into the wider AMS portal later.

Priority: Must

Acceptance criteria:

- OpenAPI docs available at /docs.
- JSON contracts stable for /nbq/next, /change-risk, /chat, /health.

US-09 - New employee

As a new employee, I want the assistant to still provide knowledge-base answers if Mistral is offline, so that onboarding is not blocked during a demo.

Priority: Should

Acceptance criteria:

- Fallback mode returns relevant CGI knowledge excerpts.
- User sees a clear message that Mistral is unavailable.

US-10 - Intake consultant

As an intake consultant, I want a test bench to try NBQ and change risk without the chatbot, so that I can validate algorithms quickly.

Priority: Should

Acceptance criteria:

- G3 test page at /g3-test allows form-based testing.
- JSON output is visible for both endpoints.

US-11 - Project referent

As a project referent, I want change risk to remain a decision-support score, so that final acceptance stays with AMS leads and not the algorithm alone.

Priority: Must

Acceptance criteria:

- Documentation states Group 3 does not own final risk sign-off.
- drivers[] provide traceability for audit.

US-12 - New employee

As a new employee on the onboarding page, I want one assistant that can answer onboarding topics AND questions about CGI risk analysis tools, so that I do not need separate apps.

Priority: Must

Acceptance criteria:

- Single chatbot on the onboarding tab.
- Backend routes /chat, /nbq/next, /change-risk, /health support the G3 engine.
- Chatbot answers cover CI/CD, security, AMS intake, change risk, and NBQ concepts.

US-13 - New employee

As a new CGI employee on the onboarding page, I want the chatbot to show the sources used to answer my question, so that I can verify the information and access the original onboarding documentation.

Priority: Must

Acceptance criteria:

- Chatbot answers display one or more source references when relevant documentation is found in the onboarding database.
- Each source reference includes at least a document title, category, and source identifier or link.
- If no reliable source is found, the chatbot clearly states that no matching documentation was available.
- The chatbot does not present unsupported information as an official CGI process.

US-14 - New employee

As a new CGI employee, I want the chatbot to clearly indicate when it is unsure or when information is missing, so that I do not rely on invented or incomplete answers during onboarding.

Priority: Must

Acceptance criteria:

- If the chatbot cannot find relevant information in the onboarding database, it explains that the answer is not available in the current knowledge base.
- The chatbot suggests checking official documentation or contacting the appropriate onboarding referent when the answer cannot be confirmed.
- The chatbot avoids inventing procedures, tools, contacts, deadlines, or internal rules.
- Uncertain answers are clearly marked as limited or incomplete.

US-15 - Knowledge manager

As a knowledge manager, I want onboarding documents stored in the relational database to be added, updated, categorized, and deactivated, so that the chatbot always relies on accurate and up-to-date information.

Priority: Should

Acceptance criteria:

- A document entry can be stored with title, content or reference, category, source, and last update date.
- Updated documents are taken into account by the chatbot knowledge retrieval process.

- Obsolete documents can be deactivated without being permanently deleted.
- The chatbot does not use deactivated documents in its answers.
- Database content can be checked or audited through document metadata.

US-16 - Developer / Group 2

As a developer, I want the chatbot backend to retrieve onboarding knowledge from the relational database, so that answers are based on structured and maintainable company documentation instead of hardcoded content.

Priority: Must

Acceptance criteria:

- The backend can query the relational onboarding database to retrieve relevant documents or document chunks.
- Retrieved knowledge includes document metadata such as title, category, source, and last update date.
- The /chat endpoint uses retrieved database content to generate onboarding answers.
- If the database is unavailable, the API returns a clear error or fallback response instead of failing silently.
- Database access and retrieval behaviour are documented in the API or setup documentation.

5. Summary Table

ID	Persona	Priority	Story (short)
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US-06	Intake consultant	Must	As an intake consultant, I want to compute a change risk score from criticality ...
US-07	AMS manager	Should	As an AMS manager, I want to see the AI pipeline status and answer mode, so that...
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			chatbot to clearly indicate when it is unsure ...
US-15	Knowledge manager	Should	As a knowledge manager, I want onboarding documents stored in the relational dat...
US-16	Developer / Group 2	Must	As a developer, I want the chatbot backend to retrieve onboarding knowledge from...

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